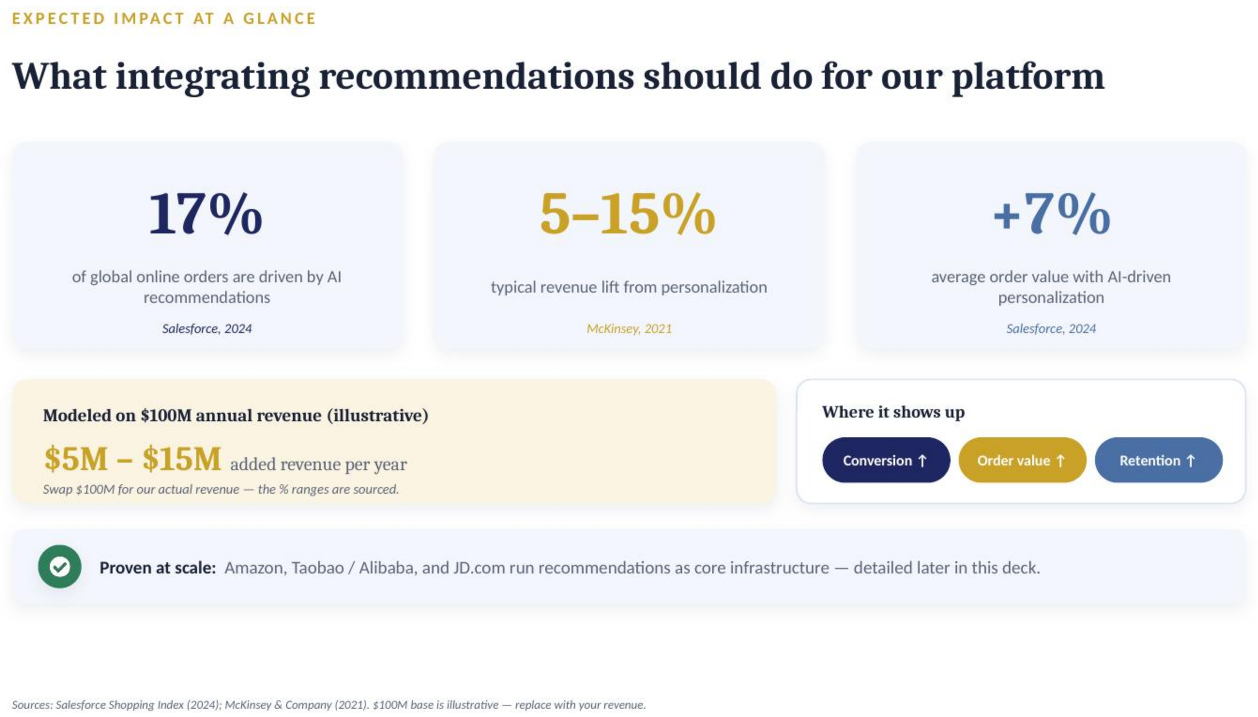
Speaker notes

Impact showcase for business stakeholders. Goal: demonstrate the projected effect of integrating a recommendation system into our existing platform, using published benchmarks. The deck opens with an at-a-glance impact summary, then supports it: why it's needed, the revenue sizing, value chain, proof at scale, SWOT, risks, a phased plan grounded in a real precedent (Amazon), and how we'd measure ROI. Every statistic is attributed to a published source, with full references on the final slide.

MODIFIABLE FOR YOUR PLATFORM

- Swap in your name and the presentation date.
- Retitle to match the specific initiative or business unit if needed.



Speaker notes

Open with the punchline so the effect lands immediately. Three benchmark stats set the ceiling and the planning band: Salesforce's 2024 Shopping Index found AI recommendations drove ~17% of global orders and lifted AOV ~7%, while McKinsey's 2021 research puts the typical revenue lift at 5–15%. Applied to a $100M revenue base, that models roughly $5–15M of added revenue a year. The KPI pills show which numbers move, and the proof line previews the case studies. Everything here is expanded and sourced in the slides that follow.

MODIFIABLE FOR YOUR PLATFORM

- Replace $100M with our actual annual revenue so the modeled uplift is ours.
- If we have a target surface (e.g., home feed) or a specific KPI baseline, add it here.

ABBREVIATIONS

- AOV = Average Order Value
- AI = Artificial Intelligence

SOURCES

- Salesforce Shopping Index, 2024 — AI drove ~17% of global orders; +7% AOV with AI/agents.
- McKinsey & Company, “The value of getting personalization right—or wrong—is multiplying,” 2021 — 5–15% revenue lift.

WHY WE NEED THIS

A bigger catalog than anyone can browse — and rising expectations

Discovery is the bottleneck

Most of our catalog is never seen; shoppers can't buy what they never find.

Choice overload stalls purchases

Endless undifferentiated options delay decisions and depress conversion.

Personalization is now expected

71% of consumers expect personalized interactions; 76% are frustrated when it's missing.

What AI recommendations drive today

17%
of global online orders driven by AI — recommendations, promotions & service

+7%
higher average order value for retailers using generative AI

Salesforce Shopping Index, 2024 (1.5B shoppers).
Recommendations are how leaders convert a huge catalog into sales.

Sources: Salesforce Shopping Index (2024); McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying" (2021).

Speaker notes

Establish the problem in business terms before any money talk. Two structural problems — discovery and choice overload — both mean lost revenue, and a third force (customer expectation) makes this urgent. The right panel quantifies the prize with recent data: Salesforce's 2024 Shopping Index (1.5 billion shoppers) found AI — including personalized recommendations — drove 17% of global online orders, and retailers using generative AI saw 7% higher average order value. The 71%/76% expectation figures are from McKinsey's 2021 personalization report.

MODIFIABLE FOR YOUR PLATFORM

- Replace the discovery framing with your platform's catalog size and the share of SKUs that get little or no impressions.
- If you have internal data on un-converted browsing sessions, add it here.

ABBREVIATIONS

- SKU = Stock-Keeping Unit (a distinct purchasable product)
- AOV = Average Order Value
- AI = Artificial Intelligence

SOURCES

- Salesforce Shopping Index, 2024 — AI drove 17% of global orders; +7% AOV for retailers using generative AI (analysis of 1.5B shoppers).
- McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying," 2021 — 71% expect personalization, 76% frustrated without it.

HOW MUCH COULD IT MOVE REVENUE?

Published lift ranges, applied transparently to our numbers

What the research shows

- Personalization most often lifts revenue 5–15% (sector range up to ~25%)
- Can reduce customer-acquisition cost by up to 50%
- Improves marketing ROI by 10–30%
- Fast-growing firms earn ~40% more of their revenue from personalization

All figures: McKinsey (2021).

Illustrative on $100M annual revenue

Applying the 5–15% revenue-lift range:

5% lift	10% lift	15% lift
$5M /year	$10M /year	$15M /year

Replace $100M with our actual revenue. The percentages are sourced; the base is a placeholder for modeling, not a forecast.

Source: McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying," 2021.

Speaker notes

This answers 'how much would it raise revenue?' honestly. We can't promise a number without our own data, so we apply McKinsey's published 5–15% revenue-lift range to a round revenue base and show the modeled uplift. Stress that the percentages are sourced and $100M is a placeholder to swap for our real revenue — this is a sizing model, not a forecast. Even at the low end, the absolute dollars usually dwarf the build cost.

MODIFIABLE FOR YOUR PLATFORM

- Replace $100M with our actual annual revenue — the tiles will then show our modeled uplift.
- If finance prefers a narrower band, use the 10–15% 'most often' figure rather than the full 5–15%.

ABBREVIATIONS

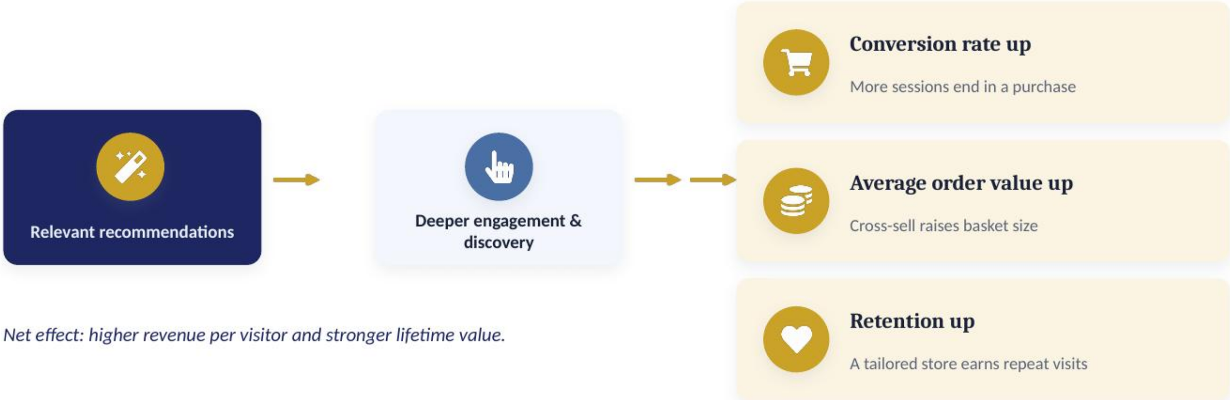
- ROI = Return on Investment
- CAC = Customer-Acquisition Cost

SOURCES

- McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying," 2021 — 5–15% revenue lift, CAC –50%, marketing ROI +10–30%, 40% more revenue from personalization.

FROM RELEVANCE TO REVENUE

How recommendations reach the numbers we report



Speaker notes

Make the causal chain explicit: relevance drives engagement and discovery, which lifts the three KPIs executives track — conversion, average order value, retention — rolling up to revenue and lifetime value. Tie each box to a metric already on our dashboard. No external statistic here; this is the mechanism behind the sourced numbers on the previous slides.

MODIFIABLE FOR YOUR PLATFORM

- Replace the three KPI boxes with the exact metrics on our executive dashboard and add current baselines.

ABBREVIATIONS

- AOV = Average Order Value (revenue ÷ orders)
- LTV = Lifetime Value (expected revenue per customer)

WHERE CUSTOMERS SEE IT

Five surfaces, each with a business job

Personalized feed

Drives engagement

“For you” home and category feeds

Frequently bought together

Grows basket size

Cross-sell at cart & product page

Trending & popular

Converts undecided shoppers

Strong default with little data

Similar items

Extends the session

“More like this” keeps browsing

Recently viewed

Recovers lost intent

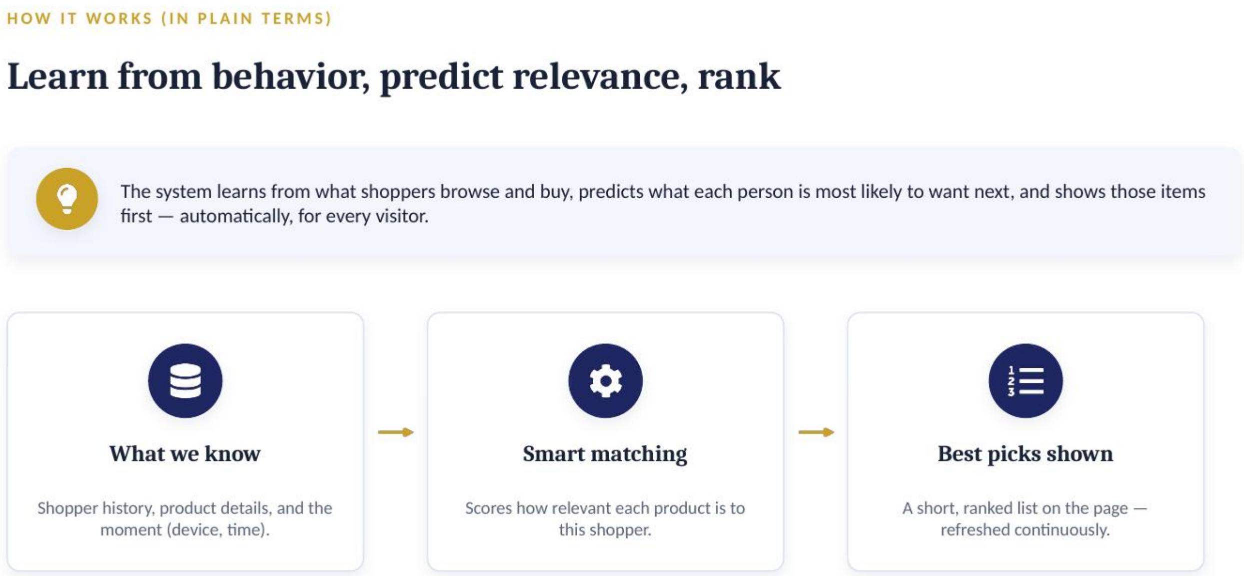
Easy path back to consideration

Speaker notes

Each surface exists to move a number. Personalized feeds deepen engagement; frequently bought together is basket-size cross-sell; trending converts shoppers you know little about; similar items extends the session; recently viewed recovers intent. Lead with the two or three surfaces that map to our biggest current gap.

ABBREVIATIONS

- FBT = Frequently Bought Together



Speaker notes

Deliberately one slide, no jargon: this is learned automation, not hand-built rules. If a stakeholder wants the mechanics, point them to the technical deck.

MODIFIABLE FOR YOUR PLATFORM

- If your audience is more hands-on, name the approach you'll start with (e.g., 'frequently bought together first').

PROOF AT SCALE

Market leaders treat recommendations as core infrastructure

Amazon

Pioneered item-to-item recommendations ("customers who bought this also bought"), launched 1998 and published in IEEE Internet Computing.

A foundational cross-sell engine, copied across the web.

Taobao / Alibaba

"Guess What You Like" home feed uses graph embeddings (EGES) to personalize a billion-item catalog; A/B-tested CTR gains.

Behavior-driven discovery at marketplace scale.

JD.com (Jingdong)

Deep-learning semantic retrieval and re-ranking serve hundreds of millions of users at ~$100B+ annual GMV.

Relevance tied directly to sales at huge scale.

Sources: Smith & Linden, IEEE Internet Computing (2017); Wang et al., KDD 2018; Li et al. (JD.com), 2021.

Speaker notes

Neutralize 'do we really need this?' Three of the largest retailers on earth treat recommendations as core infrastructure. Amazon's item-to-item approach (1998, published 2003) became the industry template. Alibaba's Taobao runs a personalized 'Guess What You Like' feed using graph embeddings (the EGES method, KDD 2018). JD.com ties deep-learning relevance to a business doing $100B+ GMV. Each claim is sourced on the references slide.

MODIFIABLE FOR YOUR PLATFORM

- If you compete with specific players, swap in the most relevant competitors and any public evidence of their recommendation efforts.

ABBREVIATIONS

- EGES = Enhanced Graph Embedding with Side information (Alibaba)
- CTR = Click-Through Rate
- GMV = Gross Merchandise Value
- JD = Jingdong (JD.com)

SOURCES

- Smith & Linden, "Two Decades of Recommender Systems at Amazon.com," IEEE Internet Computing, 2017.
- Wang et al., "Billion-scale Commodity Embedding for E-commerce Recommendation in Alibaba," KDD 2018 (arXiv:1803.02349).
- Li et al., "From Semantic Retrieval to Pairwise Ranking" (JD.com), 2021 (arXiv:2103.12982); JD scale per arXiv:1709.00300.

SWOT ANALYSIS

Adopting recommendations on our platform

Strengths

- We already own first-party behavioral data — the fuel for recommendations
- Scales personalization to every visitor automatically
- Mature, proven technology with vendor and open-source options
- Directly targets conversion, AOV, and retention

Weaknesses

- Cold start: thin signal for new users and items at launch
- Needs data quality, ML talent, and low-latency serving
- Results depend on traffic volume — thin data, weaker lift
- Ongoing maintenance and experimentation overhead

Opportunities

- 5–15% revenue lift typical (McKinsey, 2021)
- Differentiation: 71% of consumers expect personalization
- Cross-sell and surfacing of long-tail catalog
- Lower acquisition cost (up to −50%) via better retention

Threats

- Privacy regulation and consent (e.g., GDPR, CCPA)
- Done poorly, personalization can cut revenue ~5% (McKinsey)
- Competitors with mature systems raise the bar
- Bias, filter bubbles, and reputational risk if unmanaged

Source: McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying," 2021.

Speaker notes

A strategic read in one view. Strengths and weaknesses are internal (our data and capabilities); opportunities and threats are external (market and regulation). The cited figures in the bottom two quadrants are from McKinsey 2021 — the upside (5–15% lift, expectation, –50% CAC) and the downside (poorly executed personalization can reduce revenue ~5%). Use this to frame an honest go/no-go discussion.

MODIFIABLE FOR YOUR PLATFORM

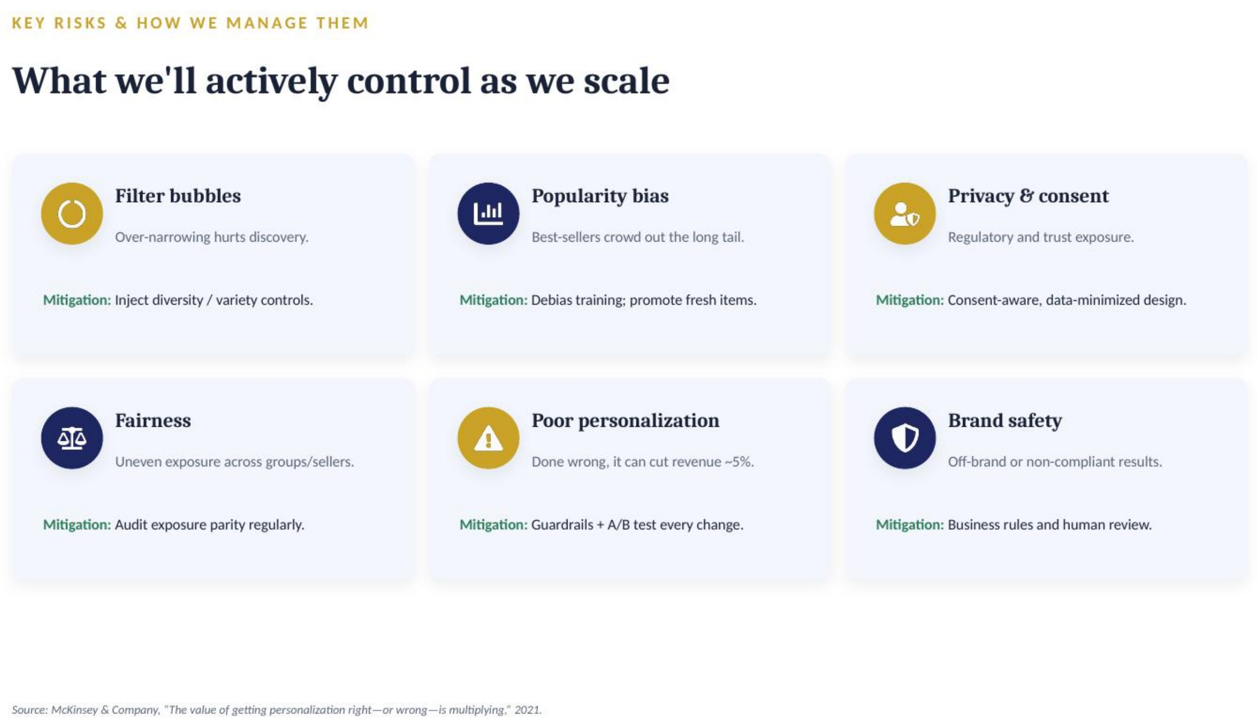
- Tailor each quadrant to our reality: our actual data assets, our talent gaps, our specific competitors, and the regulations in our markets.

ABBREVIATIONS

- SWOT = Strengths, Weaknesses, Opportunities, Threats
- GDPR = EU General Data Protection Regulation
- CCPA = California Consumer Privacy Act
- AOV = Average Order Value
- CAC = Customer-Acquisition Cost

SOURCES

- McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying," 2021 — 5–15% lift; ~5% downside if done poorly; 71% expectation; CAC up to –50%.



Speaker notes

Show the risks are known and owned, each with a concrete mitigation — what a board wants to hear. The 'poor personalization can cut revenue ~5%' figure is McKinsey's downside estimate and is why we A/B test every change behind guardrails. Frame these as responsibilities we manage, not reasons to hesitate.

MODIFIABLE FOR YOUR PLATFORM

- Add regulations specific to our markets and name the data-governance owner.
- Add any category-specific brand-safety rules we require.

ABBREVIATIONS

- A/B test = randomized controlled experiment

SOURCES

- McKinsey & Company, “The value of getting personalization right—or wrong—is multiplying,” 2021 — poorly executed personalization can reduce revenue ~5%.

A PHASED INVESTMENT

Prove value early, then scale what works

Phase 1 — Crawl
0–3 months

Popularity, trending, and rules-based “bought together.”

Quick wins, low cost

Phase 2 — Walk
3–9 months

Add behavioral collaborative filtering and similar-item recall.

Personalization begins

Phase 3 — Run
9–18 months

Personalized feeds, hybrid models, real-time ranking.

Compounding returns

Each phase ships value on its own and gates the next on measured lift. The next slide shows a real platform that followed this exact path.

Speaker notes

Phasing de-risks the investment: each phase ships value, so funding is earned by results. Phase 1 needs little new infra and creates the A/B baseline; Phase 2 adds real personalization; Phase 3 is where compounding returns live. Crucially, this isn't theoretical — the next slide shows Amazon followed precisely this crawl-walk-run path, with sources.

MODIFIABLE FOR YOUR PLATFORM

- Adjust timeframes to our team's capacity.
- Replace outcome pills with concrete targets (e.g., '+X% conversion in pilot').
- Insert budget/headcount per phase for finance.

Speaker notes

This is the evidence behind the phased plan. Amazon began with popularity and editorial curation (crawl), introduced item-to-item collaborative filtering in 1998 — computing item similarities offline for fast online lookups (walk) — published it in IEEE Internet Computing in 2003 where it proved more scalable than user-based methods, then spent two decades refining toward whole-store personalization that is continuously A/B tested (run). Both the 2017 retrospective and the 2003 paper are cited on the references slide. The takeaway: our crawl-walk-run roadmap mirrors a documented, successful path.

MODIFIABLE FOR YOUR PLATFORM

- If a competitor's published journey is more relevant to our market, substitute it here (keep the source).
- Adjust the dates/labels if you anchor the story on a different platform (e.g., Netflix's Prize-era progression).

ABBREVIATIONS

- CF = Collaborative Filtering
- A/B test = randomized controlled experiment

SOURCES

- Smith & Linden, "Two Decades of Recommender Systems at Amazon.com," IEEE Internet Computing, 2017.
- Linden, Smith & York, "Amazon.com Recommendations: Item-to-Item Collaborative Filtering," IEEE Internet Computing, 2003.

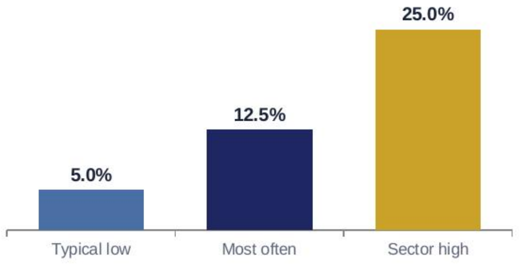
HOW WE'LL PROVE ROI

No claim ships without a controlled test

The method

1. Show recommendations to a random half of traffic (A/B test)
2. Compare against the holdout over a fixed window
3. Measure conversion, AOV, and revenue per session
4. Roll out only if lift is positive and statistically sound

What "good" looks like (published benchmark)



Personalization revenue-lift range — McKinsey (2021): most often 10–15%, sector range 5–25%.

Source: McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying," 2021.

Speaker notes

Protect credibility: we never claim a win from a dashboard moving — we prove it with a randomized A/B test against a holdout, on metrics finance recognizes. The chart shows the published benchmark range (not invented numbers): McKinsey's 5–25% sector range with 10–15% most common. We'll report our own measured lift against this benchmark once the pilot runs.

MODIFIABLE FOR YOUR PLATFORM

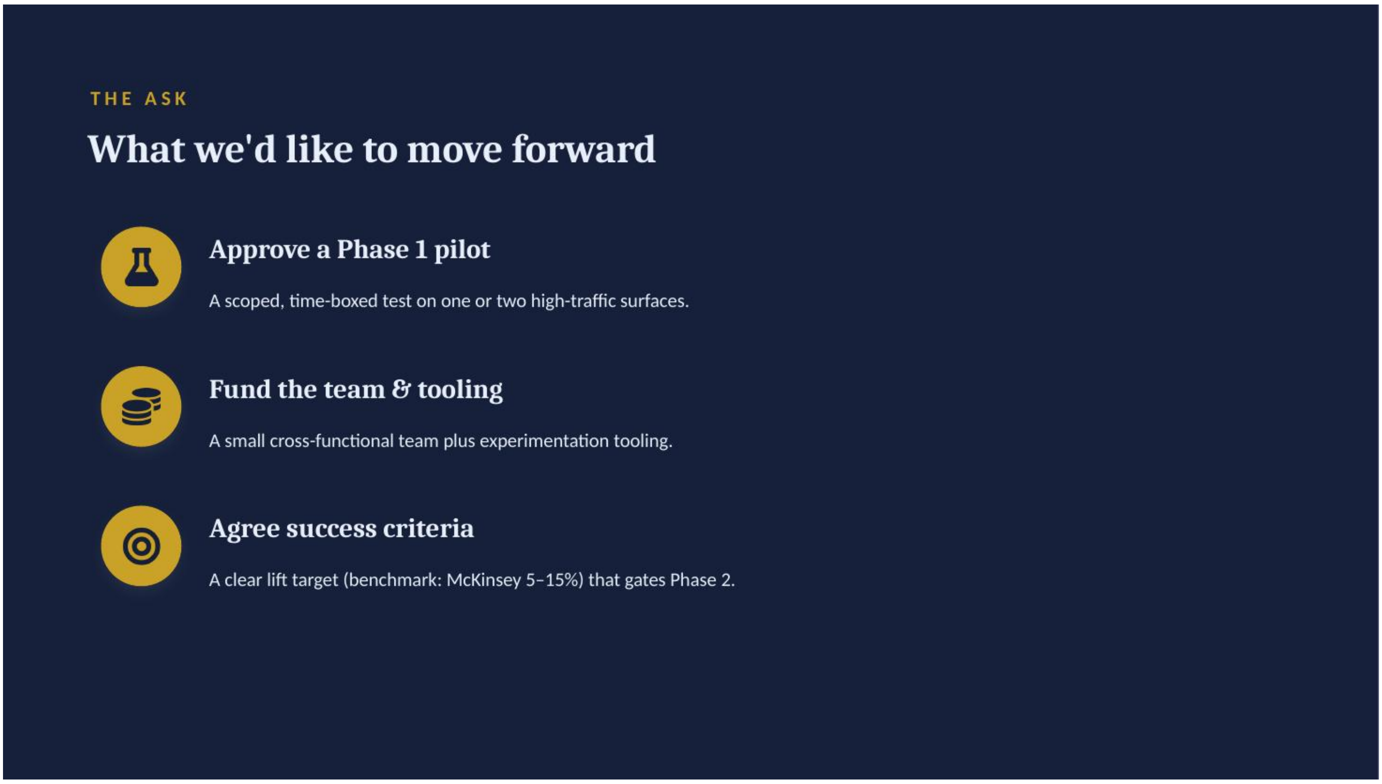
- Once the pilot runs, replace the benchmark chart with our measured A/B lift.
- Pick the primary metric finance treats as the source of truth.

ABBREVIATIONS

- A/B test = randomized controlled experiment
- AOV = Average Order Value
- CVR = Conversion Rate

SOURCES

- McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying," 2021 — revenue lift most often 10–15%, sector range 5–25%.



Speaker notes

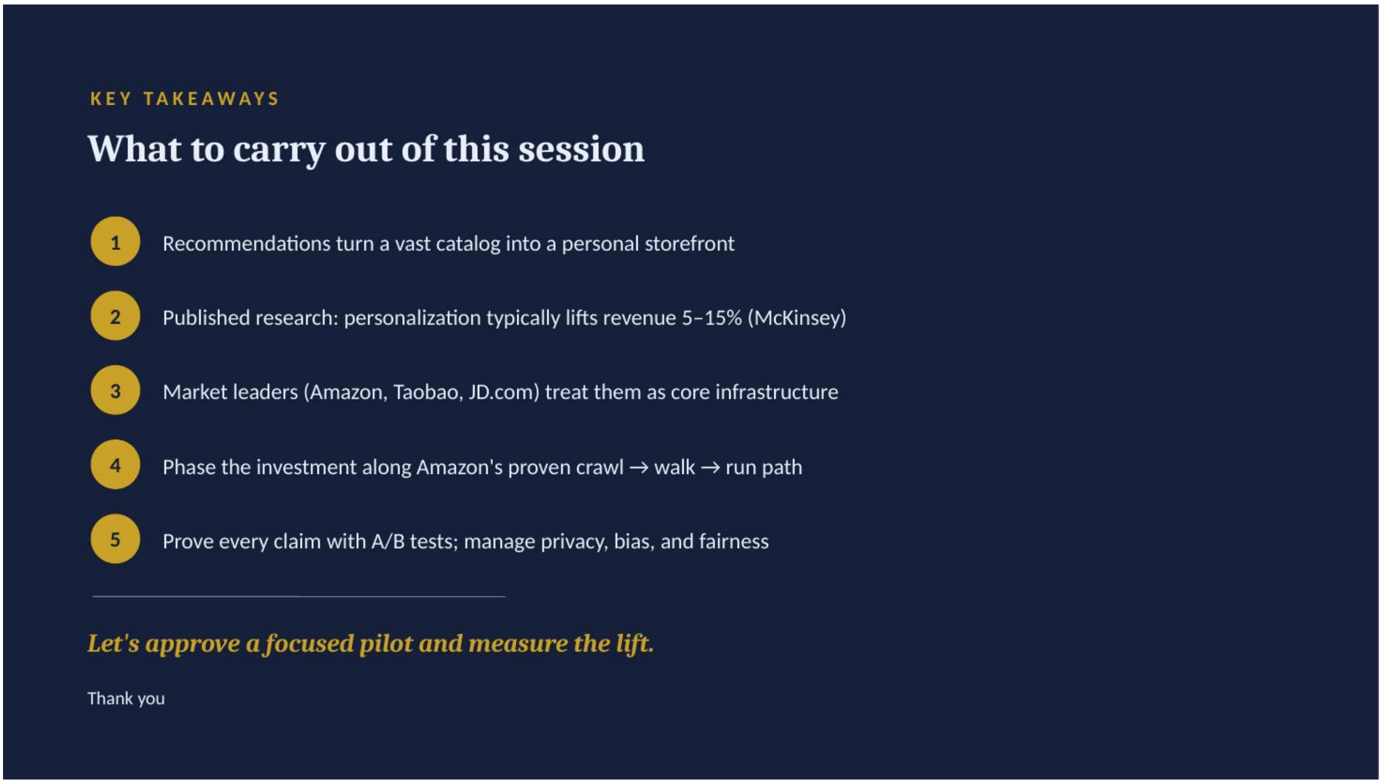
Close with a specific, decision-ready ask: a scoped pilot, a small funded team with tooling, and an agreed success metric (benchmarked to McKinsey's 5–15% lift) that gates further spend. Keep the first commitment small and measurable so it's easy to say yes.

MODIFIABLE FOR YOUR PLATFORM

- Fill in the actual pilot scope, budget, headcount, and the numeric success target.
- Name the decision-makers whose sign-off you need.

SOURCES

- McKinsey & Company, “The value of getting personalization right—or wrong—is multiplying,” 2021 — 5–15% lift benchmark.



Speaker notes

Recap and return to the ask: clear, sourced commercial value; a phased plan that mirrors a documented success; risks owned and measured. Confirm next steps and owners.

ABBREVIATIONS

- AOV = Average Order Value
- GMV = Gross Merchandise Value

SOURCES

- Full references on the next slide.

SOURCES & REFERENCES

Every figure in this deck is publicly sourced

Salesforce Shopping Index (2024)	Holiday-peak analysis of 1.5B shoppers. — AI (incl. personalized recommendations) drove 17% of global online orders; +7% AOV for retailers using generative AI. *salesforce.com/news/stories/ai-for-holiday-shopping-predictions*
McKinsey & Company (2021)	“The value of getting personalization right—or wrong—is multiplying.” — 5–15% revenue lift; CAC up to −50%; marketing ROI +10–30%; 71%/76% expectation; ~5% downside if done poorly. *mckinsey.com/capabilities/growth-marketing-and-sales/our-insights*
Gomez-Uribe & Hunt (2015)	“The Netflix Recommender System: Algorithms, Business Value, and Innovation,” ACM TMIS. — Recommendations estimated worth ~$1B/year via reduced churn. *ACM Transactions on Management Information Systems, 6(4)*
Smith & Linden (2017); Linden et al. (2003)	“Two Decades of Recommender Systems at Amazon.com” and “Amazon.com Recommendations: Item-to-Item Collaborative Filtering,” IEEE Internet Computing. *ieeexplore.ieee.org/document/7927889*
Wang et al. (2018)	“Billion-scale Commodity Embedding for E-commerce Recommendation in Alibaba,” KDD 2018 — Taobao EGES graph embeddings; A/B-tested CTR gains. *arxiv.org/abs/1803.02349*
JD.com — Li et al. (2021); Zhao et al. (2017)	“From Semantic Retrieval to Pairwise Ranking”; JD scale (hundreds of millions of users, ~$100B GMV). *arxiv.org/abs/2103.12982 · arxiv.org/abs/1709.00300*

Speaker notes

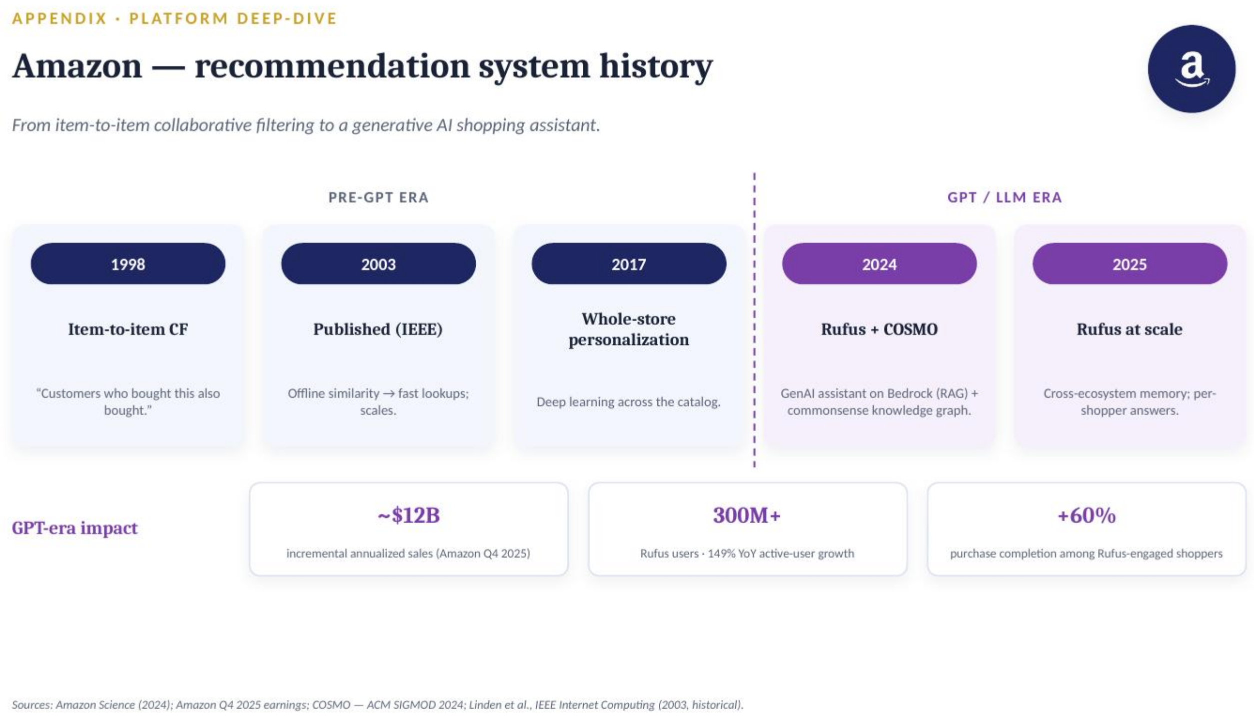
Full attribution for every statistic used in the deck. Primary sources are the Salesforce Shopping Index (2024), McKinsey's 2021 personalization report, the Netflix paper (2015), Amazon's IEEE retrospectives (2003, 2017), the Alibaba KDD 2018 EGES paper, and the JD.com papers. Offer to share the PDFs or links with stakeholders who want to verify the numbers.

SOURCES

- All URLs listed on-slide; primary/peer-reviewed and company-published sources only.

Speaker notes

Appendix divider. These slides support the deck with each platform's documented recommender history — including the post-GPT (generative/LLM) era — and effectiveness by user class. Every quantitative figure is from a source published after 2024.



Speaker notes

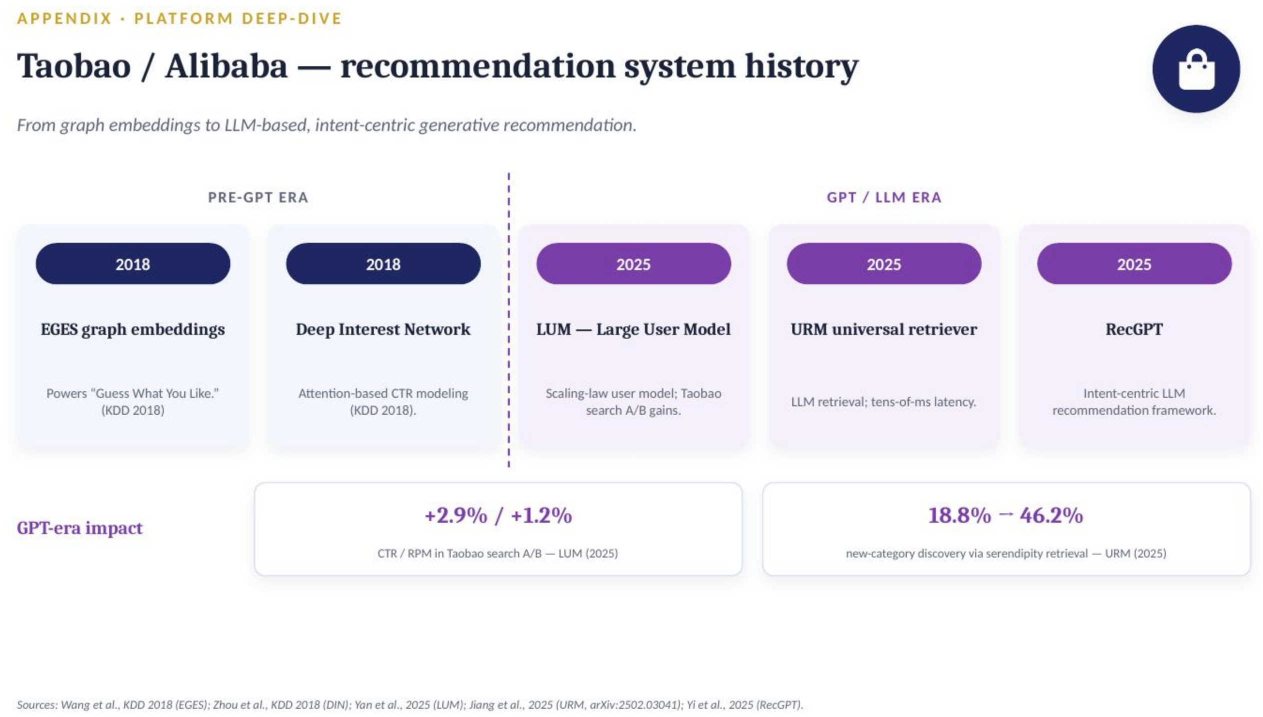
Amazon's arc: item-to-item CF (1998, published 2003) → whole-store deep-learning personalization (2017) → the LLM era with Rufus (2024) and COSMO. By 2025, Rufus reached 300M+ users and ~$12B incremental annualized sales (Amazon Q4 2025), with engaged shoppers converting ~60% better.

ABBREVIATIONS

- CF = Collaborative Filtering
- RAG = Retrieval-Augmented Generation
- GenAI = Generative AI

SOURCES

- Amazon Science (2024); Amazon Q4 2025 earnings; COSMO, ACM SIGMOD 2024; Linden et al., IEEE (2003).



Speaker notes

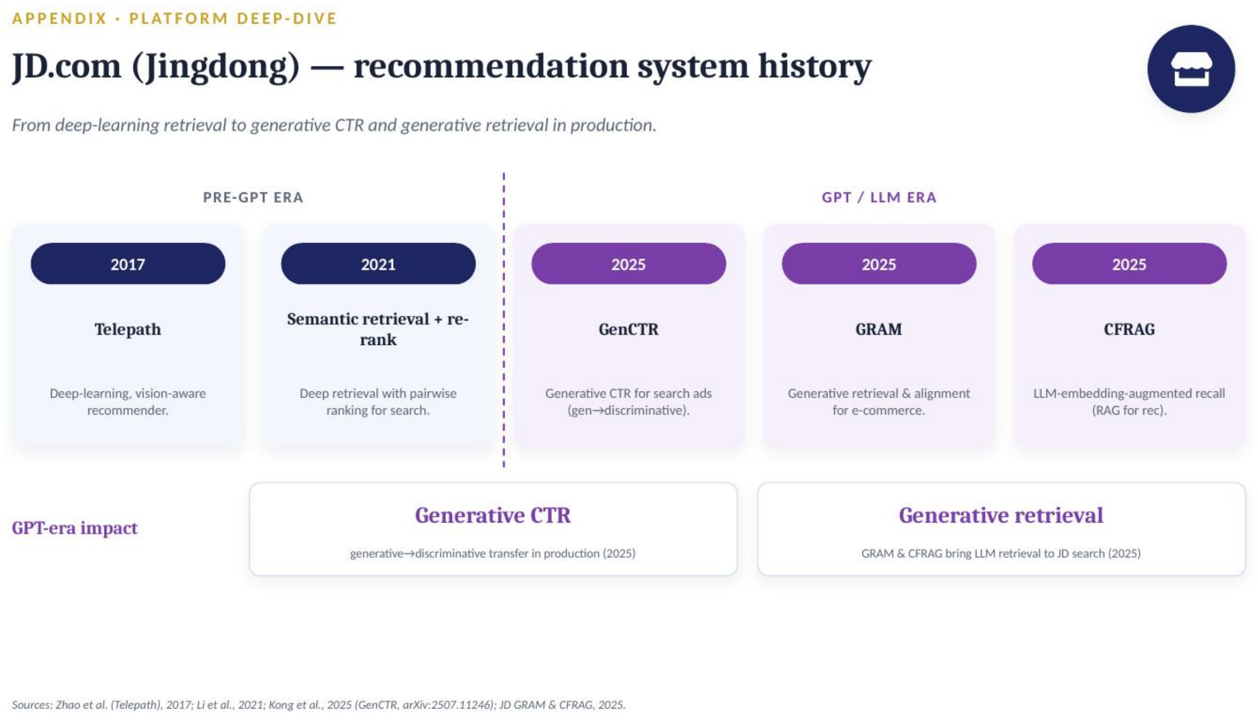
Alibaba's Taobao moved from graph embeddings (EGES, 2018) and DIN (2018) into a wave of 2025 LLM systems: LUM (+2.9% CTR/+1.2% RPM in search A/B), URM (new-category discovery 18.8%→46.2% at tens-of-ms latency), and RecGPT (intent-centric reasoning). The clearest case of a marketplace re-platforming its recommender on generative models.

ABBREVIATIONS

- EGES = Enhanced Graph Embedding w/ Side info
- DIN = Deep Interest Network
- CTR = Click-Through Rate
- RPM = Revenue Per Mille
- LUM = Large User Model
- URM = Universal Retrieval Model

SOURCES

- Yan et al., LUM (2025); Jiang et al., URM (2025, arXiv:2502.03041); Yi et al., RecGPT (2025); Wang et al., EGES (KDD 2018).



Speaker notes

JD.com progressed from deep-learning, vision-aware recommendation (Telepath, 2017) and deep semantic retrieval (2021) into 2025 generative systems (GenCTR, GRAM, CFRAG). JD publishes less segment-level data than Amazon/Alibaba, so this row is qualitative rather than headline percentages.

ABBREVIATIONS

- CTR = Click-Through Rate
- RAG = Retrieval-Augmented Generation
- GRAM = Generative Retrieval and Alignment Model

SOURCES

- Kong et al., GenCTR (2025, arXiv:2507.11246); JD GRAM & CFRAG (2025); Zhao et al. (2017); Li et al. (2021).

APPENDIX · PLATFORM DEEP-DIVE

Effectiveness by user class (LLM era)

Recent studies show the generative/LLM shift helps most where classical methods struggle — new and cold-start users.

New & cold-start

Little or no history

- **Alibaba URM:** new-category discovery 18.8% → 46.2% (2025)
- **Video platform:** +120% Recall@200 for coldest items; +36% cold-item share (2025)
- **Netflix:** +8% recall on novel-content discovery (2025)

Casual & returning

Sporadic, intent-led visits

- **Amazon Rufus:** per-shopper answers; cross-ecosystem memory (Nov 2025)
- **RecGPT (Taobao):** intent-centric reasoning replaces log-fitting (2025)
- **Conversational search:** natural-language intent matching over keywords

Core & power users

Long, rich histories

- **Amazon Rufus:** +60% purchase completion among engaged shoppers (2025)
- **Alibaba LUM:** +2.9% CTR from scaling the user model (2025)
- **Long-sequence models:** large user models exploit deep history

Sources (2025): Jiang et al. (URM); Yan et al. (LUM); Amazon Q4 2025; Li et al., Nov 2025 (Netflix); cold-start video study, Aug 2025. Platform-level per-segment figures are rarely disclosed; cross-industry studies used as directional evidence.

Speaker notes

The consistent 2025 finding: the LLM shift delivers its biggest gains for new/cold-start users, where classical CF is weakest. Casual users gain from conversational, intent-based discovery; power users feed large user models and convert far better. Caveat on the slide: the three platforms rarely publish clean per-segment numbers, so this synthesizes 2025 studies as directional evidence.

ABBREVIATIONS

- CF = Collaborative Filtering
- Recall@k = share of relevant items in top k
- CTR = Click-Through Rate
- RAG = Retrieval-Augmented Generation
- URM = Universal Retrieval Model
- LUM = Large User Model

SOURCES

- Jiang et al., URM (2025); Yan et al., LUM (2025); Amazon Q4 2025; Li et al., Nov 2025 (Netflix cold-start); cold-start video study, Aug 2025.

APPENDIX · SOURCES

LLM-era sources (2024–2026)

Amazon — Rufus	Amazon Science, "The technology behind Amazon's GenAI shopping assistant, Rufus" (2024); scale & sales via Amazon Q4 2025 earnings.	*amazon.science/blog/...-rufus*
Amazon — COSMO	Commonsense knowledge graph built with LLMs, ACM SIGMOD 2024.	*amazon.science (COSMO, SIGMOD 2024)*
Alibaba — URM	Jiang et al., "LLM as Universal Retriever in Industrial-Scale Recommender System" (2025).	*arxiv.org/abs/2502.03041*
Alibaba — LUM & RecGPT	Yan et al., "Large User Model" (2025); Yi et al., "RecGPT Technical Report" (2025).	*arXiv 2025 (LUM); RecGPT report, Aug 2025*
JD.com — GenCTR & GRAM	Kong et al., "Generative CTR Prediction..." (2025); JD "GRAM: Generative Retrieval and Alignment" (2025).	*arxiv.org/abs/2507.11246*
User-class / cold-start	Netflix cold-start (Li et al., Nov 2025); video-platform cold-start study (Aug 2025); cold-start LLM survey (2025).	*arXiv 2025*

Note: pre-GPT milestones (item-to-item CF 1998/2003; EGES & DIN 2018; JD Telepath 2017) are historical; all quantitative effectiveness figures above are from 2024–2026 publications.

Speaker notes

Consolidated citations for the appendix. All quantitative effectiveness numbers come from 2024–2026 sources; the only pre-2024 references describe historical milestones.